

第7周习题课参考内容

复合函数、隐函数微分，高阶偏导数

一、多元复合函数的求导

1. 设 $z = f(x, y)$ 在点 (a, a) 可微, $f(a, a) = a, \frac{\partial f}{\partial x}\big|_{(a,a)} = b, \frac{\partial f}{\partial y}\big|_{(a,a)} = b$.

令 $\varphi(x) = f(x, f(x, f(x, x)))$, 求 $\frac{d}{dx}\varphi^2(x)\big|_{x=a}$.

分析: 用 f'_1 和 f'_2 分别表示函数 $f(u, v)$ 对于第一个变量和第二个变量的偏导数,

理清函数的复合关系.

解: 利用复合函数微分法则求导数

$$\frac{d}{dx}\varphi^2(x) = 2\varphi(x)\frac{d}{dx}\varphi(x), \quad \frac{d}{dx}\varphi(x) = f'_1 + f'_2 \frac{d}{dx}f(x, f(x, x)),$$

$$\text{其中 } \frac{d}{dx}f(x, f(x, x)) = f'_1 + f'_2(f'_1 + f'_2),$$

$$\text{于是 } \frac{d}{dx}\varphi^2(x) = 2\varphi(x)[f'_1 + f'_2(f'_1 + f'_2(f'_1 + f'_2))],$$

将 $x = a, y = a$ 代入, 并利用题目条件

$$\varphi(a) = f(a, f(a, f(a, a))) = a, \quad f'_1(a, a) = b, \quad f'_2(a, a) = b,$$

$$\text{得到 } \frac{d}{dx}\varphi^2(x)\big|_{x=a} = 2a(b + b^2 + 2b^3). \quad \square$$

2. 设 $f(x, y)$ 是 C^1 函数, 且 $\forall x, f(x, x^2) = 1, \frac{\partial f}{\partial x}(x, x^2) = x$, 计算 $\frac{\partial f}{\partial y}(x, x^2) = ?$

解: 由题设 $f(x, x^2) = 1$, 且 $\frac{\partial f}{\partial x}(x, x^2) = x$,

$$\text{所以 } 0 = \frac{d}{dx}[f(x, x^2)] = \frac{\partial f}{\partial x}(x, x^2) + \frac{\partial f}{\partial y}(x, x^2) \frac{d(x^2)}{dx} = x + 2x \frac{\partial f}{\partial y}(x, x^2)$$

$$\text{因此 } \frac{\partial f}{\partial y}(x, x^2) = -\frac{1}{2} \quad \square$$

二、高阶偏导数

1. 设 $z = f(x + y, xy)$, 其中 $f(u, v)$ 是已知二阶连续可微函数, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

注意: 在求二阶偏导数时, 应特别注意复合函数的一阶偏导数仍然还是复合函数, 对其求导时仍要利用链式法则.

解: 令 $u = x + y, v = xy$, 则由复合函数的链式法则, 得

$$\frac{\partial z}{\partial x} = f'_u(u, v) \frac{\partial u}{\partial x} + f'_v(u, v) \frac{\partial v}{\partial x} = f'_u(u, v) + y f'_v(u, v),$$

注意到 $f'_u(u, v), f'_v(u, v)$ 仍是 x, y 的复合函数, 得

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} [f'_u(u, v) + y f'_v(u, v)] \\ &= f''_{uu}(u, v) \frac{\partial u}{\partial y} + f''_{uv}(u, v) \frac{\partial v}{\partial y} + f'_v(u, v) \frac{\partial y}{\partial y} + f''_{vu}(u, v) y \frac{\partial u}{\partial y} + f''_{vv}(u, v) y \frac{\partial v}{\partial y} \\ &= f''_{uu}(x+y, xy) + x f''_{uv}(x+y, xy) + f'_v(x+y, xy) \\ &\quad + y f''_{vu}(x+y, xy) + x y f''_{vv}(x+y, xy) \end{aligned}$$

根据题意, 式中 $f''_{uu}, f''_{uv}, f'_v, f''_{vu}, f''_{vv}$ 都是已知 2 元函数, 因此上式就是最终结果。 □

注意: 对中间变量的偏导数仍是自变量的复合函数。常见错误 $\frac{\partial}{\partial x} f'_u(u, v) = f''_{ux}(u, v)$ 。

2. 设 $z = f(xy, \frac{x}{y})$, f 的二阶偏导数连续, 求 $\frac{\partial^2 z}{\partial x^2}$ 。

解: 记 $u = xy, v = \frac{x}{y}$, $f'_1 = \frac{\partial f}{\partial u}, f'_2 = \frac{\partial f}{\partial v}$,

$$f''_{11} = \frac{\partial^2 f}{\partial u^2}, f''_{22} = \frac{\partial^2 f}{\partial v^2}, f''_{12} = \frac{\partial^2 f}{\partial u \partial v} = f''_{21} = \frac{\partial^2 f}{\partial v \partial u},$$

则
$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = y f'_1 + \frac{1}{y} f'_2,$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = y \frac{\partial f'_1}{\partial x} + \frac{1}{y} \frac{\partial f'_2}{\partial x};$$

因为 $f'_1 = \frac{\partial f}{\partial u}, f'_2 = \frac{\partial f}{\partial v}$ 都是以 u, v 为中间变量, 以 x, y 为自变量的函数, 所以

$$\frac{\partial f'_1}{\partial x} = f''_{11} \frac{\partial u}{\partial x} + f''_{12} \frac{\partial v}{\partial x} = y f''_{11} + \frac{1}{y} f''_{12},$$

$$\frac{\partial f'_2}{\partial x} = f''_{21} \frac{\partial u}{\partial x} + f''_{22} \frac{\partial v}{\partial x} = y f''_{21} + \frac{1}{y} f''_{22},$$

将以上两式代入前式得: $\frac{\partial^2 z}{\partial x^2} = y^2 f''_{11} + 2 f''_{12} + \frac{1}{y^2} f''_{22}$. □

3. 对于 2 阶连续可微函数 $u(x, y, z)$, 定义 $\Delta u := \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$, 其中

$\Delta := \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ 称为 Laplace 微分算子 (还可以定义一般 n 维 Laplace 算子)

(1) 记 $r = \sqrt{x^2 + y^2 + z^2}$, 设 $u = f(r)$, 求 $\Delta u = ?$

(2) 验证 $\Delta r = \frac{2}{r}$, $\Delta(\ln r) = \frac{1}{r^2}$, $\Delta(\frac{1}{r}) = 0$ 。【练习题 14.9: 4】

解：(1) 利用复合函数求导 $\frac{\partial u}{\partial x} = f'(r) \frac{\partial r}{\partial x} = f'(r) \frac{x}{r}$,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= f''(r) \left(\frac{x}{r}\right)^2 + f'(r) \left(\frac{1}{r} - \frac{x}{r^2} \frac{\partial r}{\partial x}\right) \\ &= f''(r) \left(\frac{x}{r}\right)^2 + f'(r) \left(\frac{1}{r} - \frac{x^2}{r^3}\right) = f''(r) \frac{x^2}{r^2} + f'(r) \frac{r^2 - x^2}{r^3}\end{aligned}$$

由 x, y, z 的对称性, 可得,

$$\frac{\partial^2 u}{\partial y^2} = f''(r) \frac{y^2}{r^2} + f'(r) \frac{r^2 - y^2}{r^3}, \quad \frac{\partial^2 u}{\partial z^2} = f''(r) \frac{z^2}{r^2} + f'(r) \frac{r^2 - z^2}{r^3}$$

$$\text{相加即得 } \Delta u = f''(r) + \frac{2}{r} f'(r);$$

(2) 特别取 $f(r) = r$, 或 $f(r) = \ln r$, 或 $f(r) = \frac{1}{r}$, 略 $\square$

推广: 导出 n 维 Laplace 算子情况下 $\Delta f(r) = ?$

4. 设 $r = \sqrt{x^2 + y^2 + z^2}$, $u = \frac{1}{r} [\varphi(r-at) + \psi(r+at)]$, 其中 $\varphi, \psi \in C^2$ 为已知函数,

求证 $\frac{\partial^2 u}{\partial t^2} = a^2 \Delta u$ 【练习题 14.9: 5】

解: 利用上题 (1) 结论, 取 $f(r) = \frac{1}{r} [\varphi(r-at) + \psi(r+at)]$ (t 看作参数)

$$\begin{aligned}\text{则 } f'(r) &= -\frac{1}{r^2} [\varphi(r-at) + \psi(r+at)] + \frac{1}{r} [\varphi'(r-at) + \psi'(r+at)] \\ f''(r) &= \frac{2}{r^3} [\varphi(r-at) + \psi(r+at)] - \frac{2}{r^2} [\varphi'(r-at) + \psi'(r+at)] \\ &\quad + \frac{1}{r} [\varphi''(r-at) + \psi''(r+at)]\end{aligned}$$

$$\text{所以 } \Delta u = f''(r) + \frac{2}{r} f'(r) = \frac{1}{r} [\varphi''(r-at) + \psi''(r+at)]$$

$$\text{另一方面 } \frac{\partial^2 u}{\partial t^2} = \frac{a^2}{r} [\varphi''(r-at) + \psi''(r+at)], \text{ 这导出需要的等式 } \square$$

三、向量值函数的微分和导数

1. 计算空间柱坐标、球坐标变换的 Jacobi 矩阵和行列式: 【练习题 14.3: 2 (2, 3)】

$$(1) \text{ 空间柱坐标变换 } \mathbf{f}(r, \theta, z) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \\ z \end{pmatrix}, \text{ 也即 } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

$$(2) \text{ 空间球坐标变换 } \mathbf{f}(r, \phi, \theta) = \begin{pmatrix} r \sin \phi \cos \theta \\ r \sin \phi \sin \theta \\ r \cos \phi \end{pmatrix}, \text{ 也即 } \begin{cases} x = r \sin \phi \cos \theta \\ y = r \sin \phi \sin \theta \\ z = r \cos \phi \end{cases}$$

解: 直接计算如下

$$(1) \quad Jf(r, \theta, z) = \begin{pmatrix} \partial x / \partial r & \partial x / \partial \theta & \partial x / \partial z \\ \partial y / \partial r & \partial y / \partial \theta & \partial y / \partial z \\ \partial z / \partial r & \partial z / \partial \theta & \partial z / \partial z \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$\det Jf(r, \theta, z) = \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r;$$

$$(2) \quad Jf(r, \phi, \theta) = \begin{pmatrix} x_r & x_\phi & x_\theta \\ y_r & y_\phi & y_\theta \\ z_r & z_\phi & z_\theta \end{pmatrix} = \begin{pmatrix} \sin \phi \cos \theta & r \cos \phi \cos \theta & -r \sin \phi \sin \theta \\ \sin \phi \sin \theta & r \cos \phi \sin \theta & r \sin \phi \cos \theta \\ \cos \phi & -r \sin \phi & 0 \end{pmatrix},$$

$$\det Jf(r, \phi, \theta) = \begin{vmatrix} \sin \phi \cos \theta & r \cos \phi \cos \theta & -r \sin \phi \sin \theta \\ \sin \phi \sin \theta & r \cos \phi \sin \theta & r \sin \phi \cos \theta \\ \cos \phi & -r \sin \phi & 0 \end{vmatrix} = r^2 \sin \phi \quad \square$$

2. 计算向量值复合函数的 Jacobi 矩阵: 【练习题 14.4: 9 (1, 3)】

(1) $f(x, y) = (x, y, x^2 y)$, $x = s + t$, $y = s^2 - t^2$, 在 $s = 2, t = 1$;

(2) $f(x, y, z) = (x^2 + y + z, 2x + y + z^2, 0)$, $x = uv^2 w^2$, $y = w^2 \sin v$, $z = u^2 e^v$ 。

解: (1) 计算每个函数的雅可比矩阵, 再利用链式法则得到复合函数的雅可比矩阵

记 $g(s, t) = (x, y)$, $x = s + t$, $y = s^2 - t^2$, 在 $s = 2, t = 1$ 时 $x = y = 3$,

$$Jf(3, 3) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2xy & x^2 \end{pmatrix}_{x=y=3} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 18 & 9 \end{pmatrix}, \quad Jg(2, 1) = \begin{pmatrix} 1 & 1 \\ 2s & -2t \end{pmatrix}_{s=2, t=1} = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix},$$

$$Jf \circ g(2, 1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 18 & 9 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & -2 \\ 54 & 0 \end{pmatrix}$$

法二: 直接计算函数复合之后的雅可比矩阵

由 $g(s, t) = (x, y) = (s + t, s^2 - t^2)$, $f(x, y) = (x, y, x^2 y)$ 得到

$$\begin{aligned} f \circ g(s, t) &= (s + t, s^2 - t^2, (s + t)^2 (s^2 - t^2)) \\ &= (s + t, s^2 - t^2, s^4 + 2s^3 t - 2st^3 - t^4), \end{aligned}$$

$$Jf \circ g(s, t) = \begin{pmatrix} 1 & 1 \\ 2s & -2t \\ 4s^3 + 6s^2 t - 2t^3 & 2s^3 - 6st^2 - 4t^3 \end{pmatrix},$$

再将 $s = 2, t = 1$ 代入即得…… (验证了法一的结果)

(2) 由题意 $g(u, v, w) = (x, y, z)$, $x = uv^2 w^2$, $y = w^2 \sin v$, $z = u^2 e^v$, 并且

$$f_1(x, y, z) = x^2 + y + z, \quad f_2(x, y, z) = 2x + y + z^2, \quad f_3(x, y, z) = 0,$$

$$Jf \circ g(u, v, w) = Jf(x, y, z) Jg(u, v, w)$$

$$\begin{aligned}
&= \begin{pmatrix} 2x & 1 & 1 \\ 2 & 1 & 2z \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v^2 w^2 & 2uvw^2 & 2uv^2 w \\ 0 & w^2 \cos v & 2w \sin v \\ 2ue^v & u^2 e^v & 0 \end{pmatrix} \\
&= \begin{pmatrix} 2xv^2 w^2 + 2ue^v & 4xuvw^2 + w^2 \cos v + u^2 e^v & 4xuv^2 w + 2w \sin v \\ 2v^2 w^2 + 4zue^v & 4uvw^2 + w^2 \cos v + 2zu^2 e^v & 4uv^2 w + 2w \sin v \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 2uv^4 w^4 + 2ue^v & 4u^2 v^3 w^4 + w^2 \cos v + u^2 e^v & 4u^2 v^4 w^3 + 2w \sin v \\ 2v^2 w^2 + 4u^3 e^{2v} & 4uvw^2 + w^2 \cos v + 2u^4 e^{2v} & 4uv^2 w + 2w \sin v \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

法二：直接计算函数复合之后的雅可比矩阵

$$\begin{aligned}
f_1 \circ g(u, v, w) &= x^2 + y + z = (uv^2 w^2)^2 + w^2 \sin v + u^2 e^v, \\
f_2 \circ g(u, v, w) &= 2x + y + z^2 = 2(uv^2 w^2) + w^2 \sin v + (u^2 e^v)^2, \\
f_3 \circ g(u, v, w) &= 0, \\
&\dots\dots
\end{aligned}$$

四. 隐函数的求导

1. 设函数 $x = x(z)$, $y = y(z)$ 由方程组 $\begin{cases} x^2 + y^2 + z^2 - 1 = 0 \\ x^2 + 2y^2 - z^2 - 1 = 0 \end{cases}$ 确定,

求 $\frac{dx}{dz}$, $\frac{dy}{dz}$, 以及 $\frac{d^2 x}{dz^2}$, $\frac{d^2 y}{dz^2}$.

解：方程组有 3 个变量，2 个方程，因而可以确定 2 个函数，有 1 个自变量。
由题意应取 z 为自变量， x, y 为函数，两方程分别关于自变量 z 求导数，得到

$$\begin{cases} 2x \frac{dx}{dz} + 2y \frac{dy}{dz} = -2z \\ 2x \frac{dx}{dz} + 4y \frac{dy}{dz} = 2z \end{cases} \quad \text{解方程组得} \quad \frac{dx}{dz} = \frac{3z}{x}, \frac{dy}{dz} = -\frac{2z}{y};$$

继续计算 $\frac{d^2 x}{dz^2} = \frac{d}{dz} \left(\frac{3z}{x} \right) = \frac{3}{x} - \frac{3z}{x^2} \frac{dx}{dz} = \frac{3}{x} - \frac{9z^2}{x^3},$

$$\frac{d^2 y}{dz^2} = \frac{d}{dz} \left(-\frac{2z}{y} \right) = -\frac{2}{y} + \frac{2z}{y^2} \frac{dy}{dz} = -\frac{2}{y} - \frac{4z^2}{y^3}. \quad \square$$

2. 设已知隐函数方程组 $\begin{cases} u = f(x, y, z, t) \\ g(y, z, t) = 0 \\ h(z, t) = 0 \end{cases}$, 其中 f, g, h 为已知光滑函数,

求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$ 以及 2 阶混合偏导数 $\frac{\partial^2 u}{\partial x \partial y}$ 。

解：变量数目分析：5（变量个数）- 3（方程个数=隐函数个数）= 2（自变量个数）

根据题意 x, y 为自变量, 因而 u, z, t 都是 x, y 的函数 (3 个方程确定 3 个隐函数);

进一步分析后面 2 个方程: 3 (变量) $- 2$ (方程) $= 1$ (自变量)

因为 y 是自变量, 所以 z, t 都是 y 的 (1 元) 函数。

由第二个方程可知 z, t 依赖于 y , 但不依赖于 x ; 这样由第一个方程导出

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x}, \quad \frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \frac{dz}{dy} + \frac{\partial f}{\partial t} \frac{dt}{dy},$$

为计算 z, t 关于 y 的导数, 后两个方程关于 y 求导得

$$\begin{cases} \frac{\partial g}{\partial y} + \frac{\partial g}{\partial z} \frac{dz}{dy} + \frac{\partial g}{\partial t} \frac{dt}{dy} = 0 \\ \frac{\partial h}{\partial z} \frac{dz}{dy} + \frac{\partial h}{\partial t} \frac{dt}{dy} = 0 \end{cases} \quad \text{解得} \quad \begin{cases} \frac{dz}{dy} = -\frac{\frac{\partial h}{\partial t} \frac{\partial g}{\partial y}}{\frac{\partial g}{\partial z} \frac{\partial h}{\partial t} - \frac{\partial g}{\partial t} \frac{\partial h}{\partial z}} \\ \frac{dt}{dy} = \frac{\frac{\partial h}{\partial z} \frac{\partial g}{\partial y}}{\frac{\partial g}{\partial z} \frac{\partial h}{\partial t} - \frac{\partial g}{\partial t} \frac{\partial h}{\partial z}} \end{cases}$$

代入前式整理得到

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x}, \quad \frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \left(\frac{\partial f}{\partial t} \frac{\partial h}{\partial z} - \frac{\partial f}{\partial z} \frac{\partial h}{\partial t} \right) \frac{\partial g}{\partial y} / \left(\frac{\partial g}{\partial z} \frac{\partial h}{\partial t} - \frac{\partial g}{\partial t} \frac{\partial h}{\partial z} \right);$$

最后 (注意只有函数 f 依赖于自变量 x , 函数 h, g 都与 x 无关)

$$\begin{aligned} \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial y} + \left(\frac{\partial f}{\partial t} \frac{\partial h}{\partial z} - \frac{\partial f}{\partial z} \frac{\partial h}{\partial t} \right) \frac{\partial g}{\partial y} / \left(\frac{\partial g}{\partial z} \frac{\partial h}{\partial t} - \frac{\partial g}{\partial t} \frac{\partial h}{\partial z} \right) \right] \\ &= \frac{\partial^2 f}{\partial x \partial y} + \left(\frac{\partial^2 f}{\partial x \partial t} \frac{\partial h}{\partial z} - \frac{\partial^2 f}{\partial x \partial z} \frac{\partial h}{\partial t} \right) \frac{\partial g}{\partial y} / \left(\frac{\partial g}{\partial z} \frac{\partial h}{\partial t} - \frac{\partial g}{\partial t} \frac{\partial h}{\partial z} \right) \end{aligned}$$

(依照题意, 函数 f, g, h 关于变量 x, y, z, t 的偏导数都是已知函数) □

五、简单偏微分方程求解

1. 解下列简单偏微分方程:

$$(1) \quad \frac{\partial^2 u}{\partial x^2} = 0; \quad (2) \quad \frac{\partial^2 u}{\partial x \partial y} = 0; \quad (3) \quad \frac{\partial^3 u}{\partial x \partial y \partial z} = 0. \quad \text{【练习题 14.9: 6】}$$

解: (1) 方程为 $\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = 0$, 所以 $\frac{\partial u}{\partial x} = f(y, z)$, 这里 $f(y, z)$ 可以取 (y, z) 的任意函数,

因而 $u = xf(y, z) + g(y, z)$, 其中 $g(y, z)$ 也可以取 (y, z) 的任意函数。

(2) 方程为 $\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = 0$, 这导出 $\frac{\partial u}{\partial y} = f(y, z)$, 这里 $f(y, z)$ 为 (y, z) 的任意函数,

$$\text{因此 } u = \int_a^y f(t, z) dt + g(x, z) = F(y, z) + g(x, z),$$

其中 $F(y, z)$ 是 (y, z) 的任意函数, 且关于 y 的偏导数存在 (并连续),

对称地, $g(x, z)$ 也是 (x, z) 的任意函数, 关于 x 的偏导数存在 (并连续)。

(3) 由方程导出 $\frac{\partial^2 u}{\partial y \partial z} = f(y, z)$, 这里 $f(y, z)$ 为 (y, z) 的任意函数,

$$\text{积分得到 } \frac{\partial u}{\partial z} = \int_a^y f(t, z) dt + g(x, z) = F(y, z) + g(x, z)$$

其中 $F(y, z)$ 是 (y, z) 的任意函数, 且关于 y 的偏导数存在 (并连续),

对称地, $g(x, z)$ 也是 (x, z) 的任意函数, 关于 x 的偏导数存在 (并连续)。

再积分得

$$u = \int_c^z F(y, t) dt + \int_c^z g(x, t) dt + W(x, y) = U(y, z) + V(x, z) + W(x, y),$$

其中 U, V, W 都是 2 次连续可微的任意二元函数。 $\square$

2. 求解偏微分方程 $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$ 【问题 14.9: 1】

解: 注意到 $x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$ 是 xy 平面上沿着原点径向方向的方向导数算子,

可以考虑极坐标代换 $x = r \cos \theta, y = r \sin \theta$, 则

$$\frac{\partial}{\partial r} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y} = \frac{1}{r} (x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y})$$

这样原方程就化为 $\frac{\partial z}{\partial r} = \frac{z}{r}$,

将方程看作关于 r 的常微分方程解得

$$z = c(\theta)r,$$

这里 $c(\theta)$ 是任意连续可微函数, 回到原自变量得到

$$z(x, y) = \tilde{c}\left(\frac{y}{x}\right) \sqrt{x^2 + y^2} \quad \square$$

注 1: 上述解也可以写成其他形式, 比如

$$z(x, y) = \tilde{c}\left(\frac{y}{x}\right) |x| \sqrt{1 + \left(\frac{y}{x}\right)^2} = \tilde{\tilde{c}}\left(\frac{y}{x}\right) x$$

注 2: 偏微分方程在直线 $x=0$ 或 $y=0$ 上退化为常微分方程, 分别有解

$$z(x, 0) = ax, \quad x \neq 0 \quad (a \text{ 为任意常数})$$

$$z(0, y) = by, \quad y \neq 0 \quad (b \text{ 为任意常数})$$

在原点退化为简单等式 $z=0$.